Physics-Informed Neural Networks for Solving Burgers' Equation: Reproducible Implementation and Analysis

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***Abstract*—Physics-Informed Neural Networks (PINNs) represent a class of deep learning models capable of solving partial differential equations (PDEs) by embedding governing physical laws directly into the loss function. This paper presents a comprehensively annotated, reproducible implementation of PINNs for solving the nonlinear 1D Burgers' equation. We formalize the theoretical foundations of both continuous (collocation-based) and discrete (Runge-Kutta time-stepping) PINN formulations. Furthermore, we analyze the network structure, training methodology, and empirical results using a continuous approach. Our ablation study demonstrates a $1490.5\times$ reduction in relative L_2 error when enforcing physical constraints compared to purely data-driven training. The complete code repository and Jupyter notebook are available at: https://github.com/ANG121999/PINNS/blob/main/pinns-project/PHYSICS_INFORMED_NEURAL_NETWORKS_(PINNs)_FOR_BURGERS'_EQUATION.ipynb**

***Index Terms*—physics-informed neural networks, PINNs, Burgers equation, Runge-Kutta, deep learning, partial differential equations, continuous-time PINN, discrete PINN, ablation study**

I. Introduction

Deep learning is increasingly utilized to solve complex scientific and engineering problems governed by partial differential equations (PDEs). Physics-Informed Neural Networks (PINNs), introduced by Raissi et al., encode physical laws directly into the neural network's loss function, enabling the model to learn solutions that are consistent with both observational data and the governing equations.

This paper presents an open-source implementation and analysis of a PINN solver for the 1D viscous Burgers' equation—a canonical nonlinear PDE fundamental to fluid mechanics, gas dynamics, and traffic flow due to its wave phenomena and shock formation. We review the theoretical distinction between continuous and discrete PINN architectures and present extensive experimental results, including an ablation study highlighting the necessity of the physics loss.

II. Governing Equations and PINN Theory

A. The Burgers' Equation

In one space dimension, the viscous Burgers' equation with Dirichlet boundary conditions is defined as:

$$u_t + uu_x - \nu u_{xx} = 0, \qquad x \in [-1, 1],\ t \in [0, 1] \quad (1)$$

where $u(t, x)$ is the latent solution and ν is the kinematic viscosity (set to $\nu = 0.01/\pi \approx 0.003183$ in our experiments). Given noisy measurements, PINNs allow us to infer the continuous spatio-temporal solution without relying on traditional dense mesh discretization.

B. Continuous Time Models

In the continuous PINN framework, we define the residual network $f(t, x)$ based on the left-hand side of the PDE:

$$f := u_t + uu_x - \nu u_{xx} \quad (2)$$

We approximate $u(t, x)$ using a deep neural network. By applying automatic differentiation, we can compute the partial derivatives required for $f(t, x)$ without numerical truncation errors. The network is trained by minimizing the composite Mean Squared Error (MSE):

$$MSE = MSE_u + MSE_f \quad (3)$$

where:

$$MSE_u = \frac{1}{N_u} \sum_{i=1}^{N_u} \left| u(t_u^i, x_u^i) - u^i \right|^2 \quad (4)$$

$$MSE_f = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| f(t_f^i, x_f^i) \right|^2 \quad (5)$$

Here, MSE_u enforces the known initial and boundary data at N_u points, while MSE_f acts as a physics-informed regularization term, enforcing the structure of the Burgers' equation at N_f randomly sampled collocation points across the spatio-temporal domain.

C. Discrete Time Models (Runge-Kutta)

An alternative to continuous sampling is the discrete time model, which leverages classical Runge-Kutta time-stepping. For the Burgers' equation, applying an implicit Runge-Kutta method with q stages yields:

$$u^{n+c_i} = u^n - \Delta t \sum_{j=1}^{q} a_{ij} \mathcal{N}[u^{n+c_j}], \quad i = 1, ..., q \quad (6)$$

where the nonlinear operator is $\mathcal{N}[u] = uu_x - \nu u_{xx}$.

Instead of globally sampling N_f collocation points, a multi-output neural network is designed to predict the solution at the intermediate stages $[u_1^n(x), ..., u_q^n(x), u_{q+1}^n(x)]$. The loss function minimizes the sum of squared errors between the network's predictions and the known data at specific time snapshots t^n. The primary advantage of the discrete PINN is its ability to employ high-order implicit schemes (e.g., $q = 500$), allowing for massive time steps (Δt) that can resolve the entire spatio-temporal dynamics in a single step without strict stability constraints.

III. Implementation Summary

Our computational implementation leverages the continuous PINN methodology to solve the Burgers' equation. The configuration is as follows:

- **Data Strategy:** $N_u = 100$ initial/boundary training points and $N_f = 10,000$ collocation points generated via Latin Hypercube Sampling.
- **Architecture:** A fully connected Multi-Layer Perceptron (MLP) consisting of an input layer (size 2), 8 hidden layers with 20 neurons each, and a single-output layer. Tanh activation functions were utilized.
- **Optimization Routine:** A two-phase training approach was employed. Initial convergence was driven by the Adam optimizer for 5,000 epochs, followed by fine-tuning using the L-BFGS optimizer for 36,063 iterations.

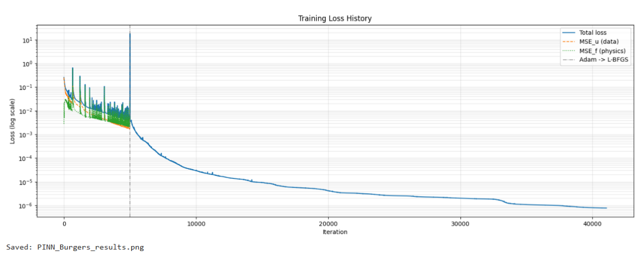

Fig. 1. Training loss history demonstrating the transition from Adam optimization to L-BFGS fine-tuning.

As seen in Fig. 1, the Adam optimizer establishes initial convergence, while the L-BFGS algorithm successfully minimizes the total loss to 7.72×10^{-7}.

IV. Results and Ablation Study

The continuous PINN demonstrated exceptional capability in capturing the complex dynamics of the Burgers' equation. After the full optimization routine, the model achieved a relative L_2 error of 5.50×10^{-4}.

Fig. 2 illustrates that the maximum absolute error remains highly localized around the shock formation, maintaining excellent accuracy throughout the rest of the domain. To further investigate the network's ability to resolve discontinuous wave phenomena, we extracted solution profiles at fixed times.

As demonstrated in Fig. 3, the PINN perfectly tracks the high-resolution exact solution, correctly modeling the nearly discontinuous internal layer without introducing numerical oscillations.

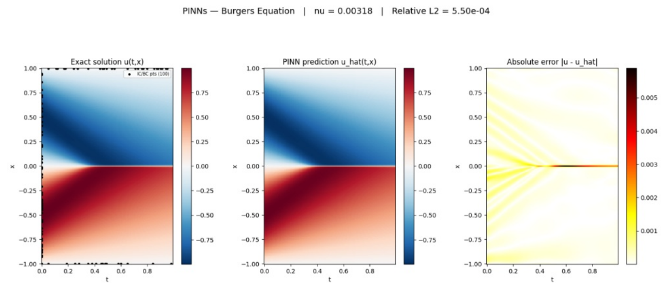

Fig. 2. Comparison of the exact solution (left), the continuous PINN prediction (center), and the absolute error magnitude (right) across the spatio-temporal domain.

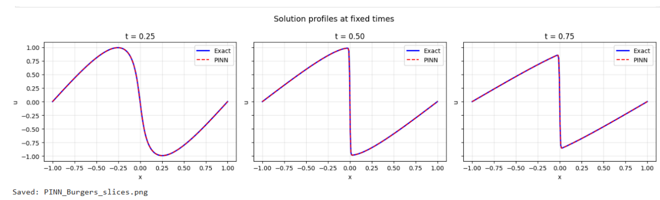

Fig. 3. Cross-sectional solution profiles at $t = 0.25$, $t = 0.50$, and $t = 0.75$, demonstrating the PINN's ability to capture sharp shock waves.

A. Ablation Study: The Impact of Physics Constraints

To rigorously quantify the value of the physics-informed constraint, we conducted an ablation study comparing the full PINN against a purely data-driven network where the PDE residual loss (MSE_f) was deactivated.

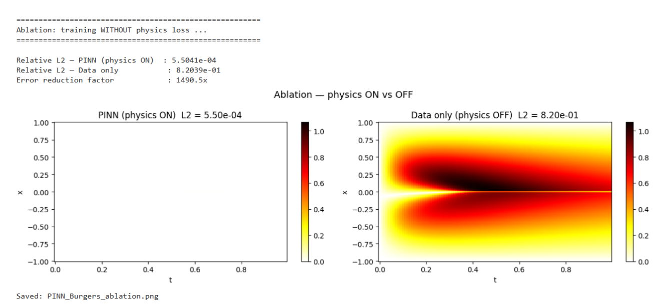

Fig. 4. Ablation study visualizing the predictive failure of a purely data-driven model (right) compared to the physics-informed model (left).

As shown in Fig. 4, removing the physics loss causes catastrophic failure in the model's predictive capability.

- **PINN (Physics ON):** Achieved a relative L_2 error of 5.50×10^{-4}.
- **Data-Only (Physics OFF):** Degraded to a relative L_2 error of 8.20×10^{-1}, resulting in a blurred, inaccurate representation of wave propagation.

Integrating the underlying physics yielded a remarkable **1490.5x error reduction factor**, explicitly validating the core hypothesis of physics-informed deep learning in small-data regimes.

V. Discussion and Conclusion

This paper documents a reproducible, open-source continuous PINN implementation for the Burgers' equation. Our results confirm that physics-informed constraints act as a powerful regularizer; utilizing only 100 scattered data points, the network synthesized the complete spatio-temporal solution with high fidelity.

While continuous PINNs offer immense sampling flexibility—freeing the solver from rigid grid dependencies—they can suffer from the "curse of dimensionality" when N_f must scale to higher dimensions. In such future scenarios, discrete PINN architectures leveraging implicit Runge-Kutta stepping present a highly stable, computationally efficient alternative.

Supplementary Materials

The presentation slides accompanying this research can be viewed here: Project Presentation.

The theoretical foundation and original discrete/continuous framework inspiring this work can be found in Raissi et al. (2017): Physics Informed Deep Learning (Part I).